# BRANCH COVERAGE REPORT

Electronics Store Inventory Management System (EIMS) · Team 9

| Test execution | Passed | Failed | Branches | Partial branches | Combined coverage |
|---|---|---|---|---|---|
| Complete pytest suite | 181 | 0 | 250 | 53 | 83% |

## 1. Objective

This report records branch-aware coverage measurement for the EIMS application. Branch coverage checks whether alternative decision outcomes (for example, true/false paths in conditional logic) are exercised by the test suite. The report is based on the pytest-cov terminal output supplied for this run.

## 2. Test Environment and Command

**Environment:** Windows 11 (win32), Python 3.12.7, pytest 8.3.5, pytest-cov 7.1.0, coverage 7.17.1, virtual environment (.venv).

**Command:** pytest --cov=app --cov-branch --cov-report=term-missing --cov-report=html:branch_htmlcov

The run collected 181 tests. All 181 passed, with one dependency deprecation warning. Execution time was 117.47 seconds.

## 3. Overall Branch-Aware Coverage

The coverage tool reported 1,139 statements, 172 missed statements, 250 branch opportunities, and 53 partially covered branch points. Its combined statement-and-branch coverage result was 83%. This combined figure is the value printed in the coverage table's Cover column; it should not be described as a separately calculated branch-only percentage.

| Source file | Stmts | Miss | Branch | BrPart | Cover |
|---|---|---|---|---|---|
| app/__init__.py | 0 | 0 | 0 | 0 | 100% |
| app/bootstrap.py | 47 | 6 | 14 | 3 | 85% |
| app/config.py | 6 | 0 | 0 | 0 | 100% |
| app/db.py | 13 | 0 | 0 | 0 | 100% |
| app/main.py | 495 | 91 | 142 | 38 | 79% |
| app/models.py | 106 | 0 | 0 | 0 | 100% |
| app/monitoring/__init__.py | 0 | 0 | 0 | 0 | 100% |
| app/monitoring/monitor.py | 47 | 47 | 6 | 0 | 0% |
| app/monitoring/service.py | 17 | 17 | 4 | 0 | 0% |
| app/repositories.py | 154 | 4 | 38 | 3 | 96% |
| app/schemas.py | 172 | 6 | 18 | 6 | 94% |
| app/security.py | 71 | 1 | 22 | 3 | 96% |
| app/services.py | 11 | 0 | 6 | 0 | 100% |
| **TOTAL** | **1,139** | **172** | **250** | **53** | **83%** |

## 4. Findings

The lowest branch-aware coverage is in app/monitoring/monitor.py and app/monitoring/service.py, both at 0%. app/main.py reports 79%, with 38 partial branch points. app/bootstrap.py reports 85%, with 3 partial branch points. app/repositories.py reports 96%, app/schemas.py 94%, and app/security.py 96%.

The detailed terminal output lists several uncovered decision transitions in app/main.py, including transitions displayed in the form line->line. The generated HTML report in branch_htmlcov provides the detailed per-file view and should be retained as supporting evidence.

## 5. Test Execution Result

All 181 tests passed. No failed or skipped tests were reported. One Starlette DeprecationWarning was emitted because anyio.abc.BlockingPortal is deprecated; the warning did not cause a test failure.

## 6. Limitations and Interpretation

The 83% value is the combined coverage percentage reported by coverage.py when branch measurement is enabled. The terminal summary does not provide a separate branch-only percentage, so none is asserted here. A branch count or partial-branch count is not itself a branch percentage. Coverage is limited to the app package selected by --cov=app and to this test run/environment.

## 7. Conclusion

The full EIMS pytest suite passed all 181 tests in the branch-measurement run. The branch-aware combined coverage was 83%, with 250 branch opportunities and 53 partially covered branch points. The monitoring modules and decision paths in app/main.py are priority candidates for additional meaningful tests, where those behaviours are required by the application.